

NAIROBI SCHOOL 2025 PREDICTION



KCSE ANTICIPATION EXAM



232/3

PHYSICS

PAPER 3 (PRACTICAL)

TIME: 2½ HOURS

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

(Kenya Certificate of Secondary Education)

INSTRUCTIONS TO CANDIDATES:-

- Write your **name**, **index number** and **school** in the spaces provided above.
- Answer **all** the questions in the spaces provided in the question paper
- You are supposed to spend the first **15 minutes** of 2 ½ hours reading the whole paper carefully before commencing your work.
- Marks are given for a clear record of the observations actually made, their suitability, accuracy and the use of them
- Record your observations as soon as you make them.
- Mathematical tables, slide rules and silent non-programmable electronic calculators may be used. Take $g = 10\text{ms}^{-2}$

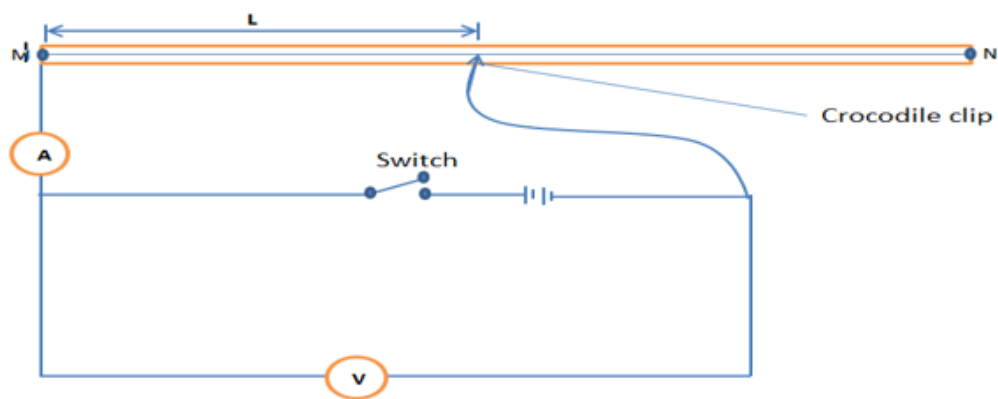
FOR EXAMINERS' USE ONLY:

QUESTION	MAXIMUM SCORE	CANDIDATE SCORE
1	20	
2	20	
TOTAL SCORE	40	

QUESTION 1

- a) You are provided with the following apparatus:
- Resistance wire fitted on a millimeter scale labeled MN
 - Switch
 - Voltmeter (0-3v) or(0-5v)
 - Ammeter (0-2.5) or(0-3)
 - Two dry cells
 - Six connecting wires

i) Set up the apparatus as shown in the fig 1 below.



ii) Remove the crocodile clip from the resistance wire MN and close the switch. Record the voltmeter reading.

$V_0 = \dots\dots\dots V$ (1mk)

iii) Attach the crocodile clip to the resistance wire such that $L=0$.

Read and record the ammeter reading at $L=0$

$I_0 = \dots\dots\dots A$ (1mk)

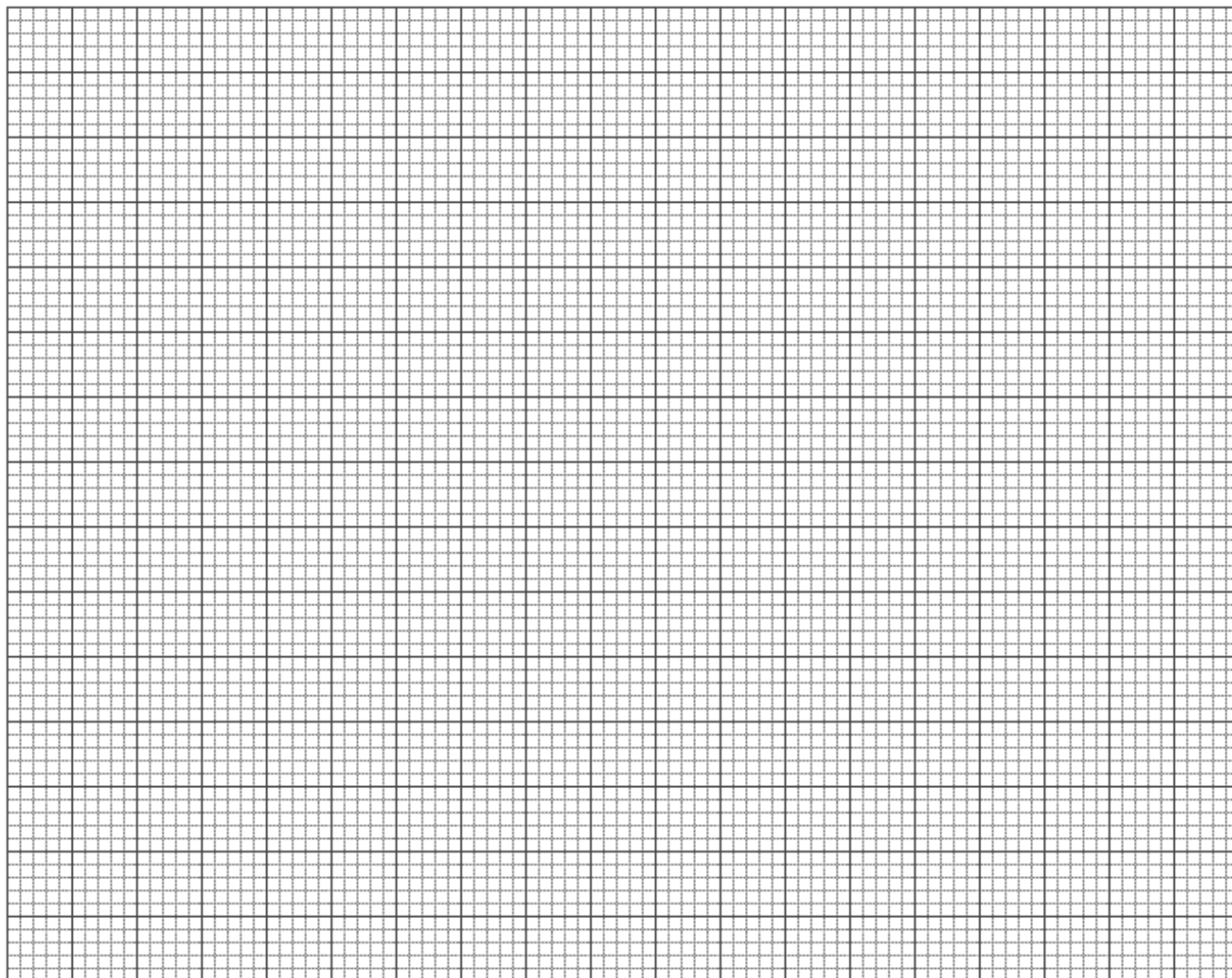
iv) Repeat the procedure in (iii) and for $L=10\text{cm}$, 20 cm , 30 cm , 40 cm , 50cm , 60 cm , 70cm and 80 cm and record the voltmeter and ammeter reading in the table below.

v) Complete the table below. (7marks)

Length L(cm)	10	20	30	40	50	60	70	80
Current I(A)								
p.d V(V)								
$V_0 - V$								
$\frac{V}{V_0 - V}$								

$\frac{V}{I} = R (\Omega)$								
----------------------------	--	--	--	--	--	--	--	--

vi) a) Plot the graph of $\frac{V}{V_0 - V}$, vertical axis against R **(5marks)**



b) Determine the slope of the graph **(3marks)**

c) Given that $\frac{v}{v_0 - v} = \frac{MR}{5} + d$

Where M and d are constants, Determine the values of M and d from the graph. **(3marks)**

QUESTION 2

PART A

You are provided with the following.

- A concave mirror
- A mirror holder.
- A screen
- A meter rule
- A candle
- A match box (to be shared)

Proceed as follows:

- a. i) Using the concave mirror provided focus a sharp image of a distant object (a laboratory window) on the screen. Measure the distance (X) between the mirror and the screen at which a sharp image is obtained repeat this three times, and record your readings in the table below. **3mks**

TRIAL	Distance(X) (cm)
1	
2	
3	

- ii) Calculate the average value of X. **(2 marks)**

iii) What is the physical significance of the result obtained in (ii) above? (1 mark)

b) Set up the apparatus as in figure 2 below.

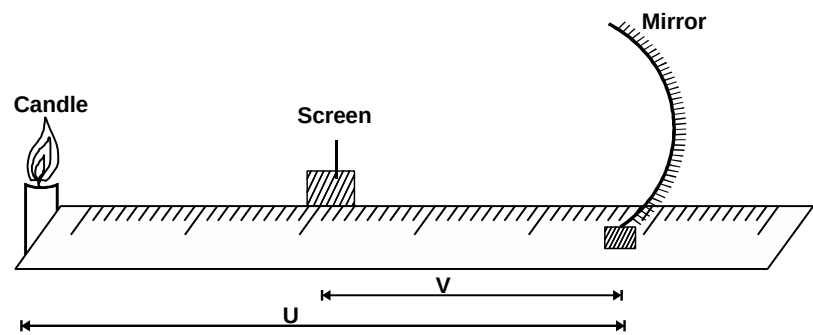


Fig 2

- c) Put the object at a distance $u = 30\text{cm}$ and from the concave mirror. Adjust the position of the screen until a sharp image is formed on the screen. Record the distance V
- d) Repeat procedure (c) above for the distance $u = 40\text{cm}$ and record the new distance V , in the table below
- e) complete the table below.

$U(\text{cm})$	$V \text{ (cm)}$	$m =V/U$	$(m + 1)$
30			
40			

3marks

f) Given that ,

calculate the values of f in each case; (3 marks)

2 **PART B**

You are provided with the following :

- a boiling tube
- a measuring cylinder
- a half metre rule
- water in a container
- a stand complete with boss and clamp
- vernier callipers (may be shared)

Proceed as follows:

- a) i)** Using the vernier callipers measure the internal diameter,(d),of the boiling tube provided

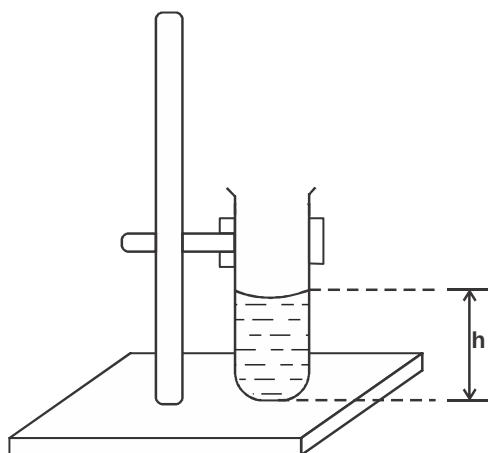
d =..... cm **(1/2 mark)**

- ii)** Using the half meter rule measure the height of the boiling tube provided

h=.....cm **(1/2 mark)**

- iii)** Calculate the volume of boiling tube **(2marks)**

- b) i)** Clamp the boiling tube vertically as shown in the figure 1 below.



ii. Using the measuring cylinder pour 25cm^3 of water into the boiling tube. Measure and record in the table below the height h , of water in the boiling tube.

iii) Repeat the procedure in b (ii) for other volumes of water, V , shown in the table.

3marks

Volume, V , of water (cm^3)	Height, h , of water (cm)	$S = V/h$
25		
35		
45		

iv) Calculate the average value of S and state what it represents

2marks